

CSE200: Online-1 on LaTeX (B1)

Instructions

- Reproduce the following article using $\text{\LaTeX}$ exactly as it is.
- You do not need to reproduce this instruction page.
- Ensure that all text formatting, lists, tables, equations, figures, references are implemented as presented in the article with the appropriate LaTeX commands.

Mark Distribution

Component	Marks
Text Formatting	15
Lists	15
Equations	25
Tables	25
Figures	20
Total	100

CSE200: Online-1 on LaTeX

Your Name (Student ID)

January 11, 2026

1 Introduction

High-quality technical writing relies on **accuracy** and *organized presentation* to convey complex concepts clearly. In this example, we show how to combine **text**, *tables*, formulas, and graphics to create a coherent document.

Employing **distinct section titles**, *well-structured lists*, and key phrases helps the reader follow the content effortlessly. Furthermore, careful use of font variations, inline expressions (y_i, z_j), and deliberate spacing enhances the professional appearance of the document.

2 Key Concepts

2.1 Advanced Nested and Mixed Lists

1. System Initialization

- (a) Software Setup
 - Install main packages
 - * Configure environment variables
- (b) Hardware Setup

2. Data Handling

- (a) Input Preparation
- (b) Feature Engineering
 - i. a) Generate new features

- ii. b) Encode categorical data
- iii. c) Normalize values
 - Check ranges

3. Evaluation and Reporting

- * Identify anomalies
- + Provide recommendations
- Output Formatting
 - * Export CSV
 - Generate summary charts

3 Mathematical Expressions

Mathematical modeling is essential for analyzing various phenomena. Consider the following examples:

$$P = \frac{1}{2}CV^2 + RI^2 \tag{1}$$

$$\begin{aligned} f(t) &= \cos(t^2) + \ln(t + 3) \\ &= \sqrt{\cos^2(t^2) + \ln^2(t + 3)} \end{aligned} \tag{2}$$

$$J = \int_1^4 (3x^2 - 2x + 5)dx$$

Piecewise function:

$$g(y) = \begin{cases} \frac{1}{y+1} & \text{if } y < 1 \\ \sqrt{y^3 + 2} & \text{if } y \geq 1 \end{cases}$$

$$\begin{aligned} z(x) &= \sum_{n=1}^{\infty} \frac{x^n}{n^2} - \frac{\ln(x+1)}{x+2} \\ &\quad + \sqrt{x^2 + 4} - \sin x \end{aligned} \tag{3}$$

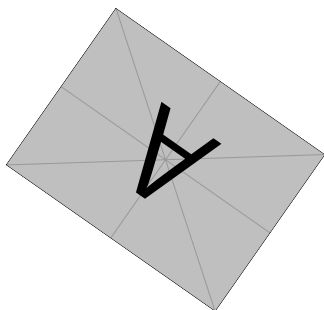
4 Evaluation Summary

Phase	Component	Score	Outcome
Phase I	Unit A	68	Review
	Unit B	74	Acceptable
	Unit C	81	Approved
Phase II	Unit A	85	Approved
	Unit B	79	Acceptable
Final Assessment		82	Pass

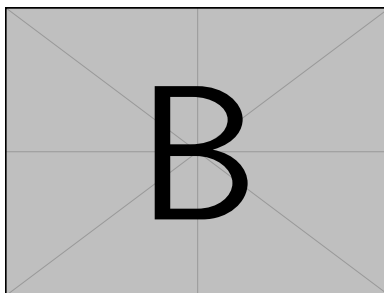
Table 1: Phase-wise evaluation with grouped rows and summary

Table 1 summarizes the performance of different components across multiple evaluation phases. The final assessment row provides an overall result based on aggregated scores.

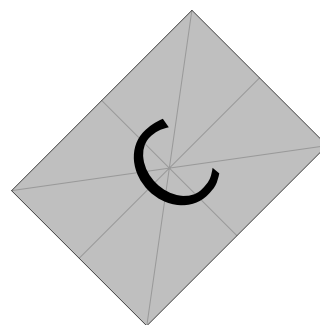
5 Figures



(a) Left-aligned



(b) Center-aligned



(c) Right-aligned

Figure 1: Three-row subfigure layout using for alignment

Figure 1 demonstrates left, center, and right alignment using subfigures with , along with scaling and rotation.

6 Conclusion

This document demonstrates **structured technical writing** using L^AT_EX with *diverse examples*, including tables, figures, and equations. Students should pay attention to consistent layout, **accurate alignment**, and *correct formatting* to achieve top marks.

These extra lines give further clarification and guidance for understanding the material. They are plain text presented without any additional formatting.